EpiSpace: Counterfactual Episode Families for Systematic Spatial Composition

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Abstract

Spatial QA accuracy alone cannot distinguish a reusable scene model from a question-specific shortcut. We ask a narrower causal question: under matched facts and accounted visual compute, does observe-first, read-many supervision improve unseen compositions over isolated QA? We introduce EpiSpace, a geometry-to-episode compiler whose basic unit is a counterfactual sibling family. Siblings share a scene and latent fact while a single evidence or reference-frame intervention requires the answer to change, remain valid, or become unknown. Typed answer-free programs define a signature-disjoint composition split; geometry remains hidden and is used only to construct and independently replay certificates. The frozen pilot audits 322 simulator jobs, retains 193 strict bundles over 46 scenes, and extracts 1063 raw RGB views from 172 write trajectories. It compiles 1093 questions and 133 complete families—91 evidence triples, 37 claim pairs, and 5 frame-equivariant pairs. The release exports 651 exactly paired training facts and a 115-item core benchmark containing 42 scene-disjoint families and 27 held-out-program items. All 1093 targets replay from source geometry; raw-RGB purity, family completeness, input–target functionality, split locks, and target-view hash isolation pass. We also release deterministic fact-matched and image-occurrence-matched schedules and define accuracy-conditioned evidence sensitivity, which cannot be maximized by always abstaining. This paper reports a validated data asset and a falsifiable training protocol; no model improvement is claimed before the frozen comparison is run.

1 Introduction

Multimodal large language models (MLLMs) can recognize furniture yet fail when the answer requires joining separated views, changing reference frames, remembering an object that left view, or distinguishing unseen from absent. VSI-Bench, MindCube, MMSI-Bench, and ViewSpatial-Bench expose complementary versions of this gap (Yang et al., 2024; Wang et al., 2026; Yang et al., 2026; Li et al., 2025). Large instruction collections such as SenseNova-SI broaden coverage, but their analyses also show shortcut sensitivity and mixed gains from textual chain-of-thought recipes (Cai et al., 2026). The remaining issue is not simply how many spatial answers a model sees, but which computation its supervision makes economical.

An isolated sample (O, q, y) does not identify that computation. A model may integrate the ordered observations O into a reusable belief and execute the query q, or map local image–language cues directly to y. Answer loss rewards both. A fluent rationale does not establish that its intermediate claims were correct or causally used. Explicit maps, tools, simulator QA, and streaming-memory benchmarks already provide valuable alternatives (Zhi et al., 2026; Wei et al., 2026; Gwak et al., 2026; Dai et al., 2026); our goal is therefore not to claim another first instance of any one component. We instead seek a controlled data intervention that can distinguish reusable spatial composition from per-question fitting.

Our first-principles unit is a counterfactual episode family. The scene and latent fact are fixed across sibling records, while one legal intervention changes the available evidence, the queried frame, or the truth of a claim. A complete evidence family contains (i) a matched prefix lacking the decisive visual evidence, (ii) a revealed sibling containing it, and (iii) a sibling in which that evidence is replaced; the answer must follow unknown → known → unknown. Frame siblings hold the two RGB images fixed while changing which camera defines forward, so the categorical direction must transform. Claim siblings use the same cross-view geometry but balance true and false propositions. These obligations make evidence sensitivity measurable and expose the always-guess and always-abstain solutions.

We combine families with an observe-first, read-many training format. An ordered RGB stream is presented before any downstream question, after which up to six heterogeneous reads share that observation. The model-visible interface remains ordinary RGB and language. A paired isolated-QA arm is compiled from the identical fact and image set. We provide two explicit comparison regimes: fact matching measures observation amortization, whereas an importance-weighted schedule repeats episode records until both arms have identical image occurrences and pixel exposure. Thus "same supervision" and "same compute" are separate, testable contracts rather than informal claims.

EpiSpace realizes this design as a read-only compiler over OmniGibson/BEHAVIOR-1K acquisitions (Li et al., 2024). It extracts raw RGB from sensor archives; rejects trajectory–question mismatches; resolves entity references with instance masks and a minimum-pixel rule; builds an observable anchor-camera belief containing only seen entities; and attaches a typed, answer-free program over grounding, frame, belief, metric, relation, perspective, and verification operations. Programs, poses, depth, masks, and certificates never enter the primary prompt. Each target is independently recomputed from the source bundle, and a target render used for perspective verification is excluded by both path and RGB hash.

The frozen pilot turns 322 planned jobs into 193 strict bundles over 46 scenes. It contains 1063 model-visible RGB views, 1093 replayed questions, and 133 complete sibling families. The paired train export supervises 651 facts; the core benchmark contains 115 records, including 42 scene-disjoint families and 27 instances of two semantic program signatures absent from training although every constituent operation is seen.

Our contributions are:

- A falsifiable supervision unit. Counterfactual evidence, frame, and claim siblings turn persistence, transformation, and epistemic honesty into joint obligations rather than independent QA.
- A fail-closed geometry-to-episode compiler. Exact-input certificates, raw-channel isolation, typed hidden programs, reference-resolution gates, scene/family locks, and independent target replay produce auditable RGB–language records without exposing an oracle scaffold to the student.
- A controlled benchmark and training contract. Same-fact and image-occurrence-matched schedules isolate information structure; semantic-signature holdout tests composition; accuracy-conditioned evidence sensitivity prevents consistent-but-wrong behavior from appearing successful.

This submission reports the frozen data asset, its independent corpus audit, and deterministic comparison schedules. Model-training, external-transfer, and human-evaluation results are intentionally outside the present evidence and are not claimed.

2 Related Work

Spatial supervision and evaluation. SpatialVLM scales metric supervision from reconstructed 3D cues (Chen et al., 2024), while SenseNova-SI spans eight million examples and explicitly studies overfitting, language shortcuts, and preliminary spatial chain-of-thought recipes (Cai et al., 2026). Multi-view and video benchmarks expose complementary deficits: VSI-Bench tests recall from scene walkthroughs (Yang et al., 2024), MindCube probes mapping, perspective-taking, and mental simulation from sparse views (Wang et al., 2026), MMSI-Bench supplies expert-written multi-image questions and stepwise error diagnoses (Yang et al., 2026), and ViewSpatial-Bench varies the queried reference frame (Li et al., 2025). ReVSI shows that a nominally correct 3D label may still be invalid for the frames actually shown to a VLM, and therefore audits answerability at each frame budget (Zhang et al., 2026b). ViewDiag further shows that cross-view agreement can coexist with metric error—consistency alone can measure evidence-insensitive collapse rather than geometry (Bhat & Yamasaki, 2026). These findings motivate our two safeguards: every item is certified against its exact RGB inputs, and family consistency is reported only together with atomic accuracy and response to decisive evidence.

Programmatic and episodic data. SPRITE already establishes simulator-grounded spatial QA at scale: language models write executable programs that are checked against simulator metadata, yielding 300K+ instruction pairs from 11K+ scenes (Zhi et al., 2026). S3-Bench similarly couples simulated and real streaming trajectories with time-grounded queries and active exploration, and AMF-VLM adds bounded-memory folding (Wei et al., 2026). SpaMEM evaluates action-conditioned belief updates and long-horizon reconstruction in dynamic houses (Liao et al., 2026); EMemBench generates verifiable questions from each agent's own game trajectory (Li et al., 2026a). Accordingly, we do not claim the first simulator QA generator, executable spatial program, or trajectory benchmark. Our unit of supervision is instead a counterfactual sibling family: matched inputs preserve, transform, reveal, delete, or leave insufficient the evidence for one latent fact. The family supplies relational obligations that isolated QA collections do not identify.

Spatial maps, memories, and tools. Cog3DMap recurrently grounds semantic tokens in an explicit 3D memory (Gwak et al., 2026); S-Agent accumulates scene and agent memory through a hierarchy of spatial tools and distills successful tool trajectories (Dai et al., 2026); and OmniView-Space reconstructs geometry, re-anchors it into query-aligned cognitive maps, and distills egocentric reasoning (Li et al., 2026b). Theory of Space instead evaluates active exploration, belief probing, and revision under false beliefs (Zhang et al., 2026a). These works already occupy claims of explicit maps, persistent memory, tool use, and active belief construction. EpiSpace is an orthogonal training-data experiment: the student receives RGB and language, whereas simulator state and typed programs remain hidden certificates used for filtering, splitting, and scoring. This follows the broader lesson of Visual Program Distillation that executed structure can supervise a single-pass VLM (Hu et al., 2024), but does not require the student to emit a program or call a geometry tool at inference time.

Precise scope of the contribution. Our claim is the conjunction of four controls rather than any component in isolation: (i) certified sibling families with evidence-change, unknown, frame-equivariant, and balanced-claim obligations; (ii) a semantic-program-signature hold-out in which every atomic operation is observed during training but selected compositions are not; (iii) a paired, same-fact and compute-accounted comparison of isolated QA against query-agnostic observe-first, read-many training; and (iv) accuracy-conditioned evidence sensitivity. Together these controls ask a narrower causal question than prior benchmarks: does episode structure, rather than extra facts, extra pixels, or an external map, improve systematic spatial composition?

3 A First-Principles Learning Contract

Why isolated answer supervision is insufficient. Let $O_{1:M}$ be an ordered observation stream and q_k one of several possible spatial reads. Isolated SFT minimizes $-\log p_\theta(y_k \mid O_{1:M}, q_k)$. This objective assigns equal reward to a reusable write–read solution

$$S_M = W_\theta(O_{1:M}), \qquad \hat{y}_k = R_\theta(S_M, q_k), \tag{1}$$

and to query-specific functions $\hat{y}_k = H_{\theta,k}(O_{1:M}, q_k)$. More paraphrases do not remove this non-identifiability; a text rationale may itself be another unverified target. Our contract changes both the presentation of evidence and the relations among targets.

Observe first, then reuse. In the episode arm, all RGB observations precede a batch of at most six heterogeneous questions. Perception is therefore not primed by a known query, and a single observation can support grounding, memory, relation, metric, perspective, and verification reads. The isolated arm contains the identical selected facts and the same per-fact view sets but repeats the observation for each question. Fact matching and compute matching are intentionally distinct: the former exposes the amortization advantage; the latter uses a logged sampler and inverse-repeat loss weights to equalize image and pixel occurrences.

Families turn evidence into an intervention. For a latent fact z_f, a family contains siblings $\{(O_f^{(v)}, q_f^{(v)}, y_f^{(v)})\}_{v\in\mathcal{V}_f}$ generated by legal interventions I_v. The pilot materializes three obligation types.

- Evidence triples share the question and image count. Two context images are held fixed for cross-view triples (one for presence triples); replacing only the decisive image changes unknown→known→unknown.
- Frame pairs share exactly the same two RGB images and entity pair. Only the camera declared as the forward frame changes; the relation label must transform accordingly.
- Claim pairs share the reconstructed cross-view relation while balancing a contradicted and a geometrically correct proposition.

Every sibling shares one consistency and split identifier. A family is discarded if a required member, unique reference, or matched-input obligation cannot be certified.

Accuracy-conditioned evidence sensitivity. Let $c_{f,v} \in \{0, 1\}$ indicate atomic correctness. Family exact match is $\prod_{v\in\mathcal{V}_f} c_{f,v}$. For evidence triples with prefix p, reveal r, and deletion d, we report reveal accuracy separately and

$$\mathrm{ES}_{\text{correct}} = \frac{\sum_f c_{f,r}c_{f,p}c_{f,d}}{\sum_f c_{f,r}}, \tag{2}$$

with an undefined denominator reported as N/A, never as a perfect score. Always abstaining makes $c_{f,r} = 0$ and therefore cannot score; always guessing fails the two unknown siblings. Frame-pair and claim-pair exact match are reported separately. Confidence intervals resample scenes, not nominal QA rows.

A safe canonical observable belief. The compiler may construct S_M as auxiliary supervision, but never exposes omniscient world state. Its origin is the ground projection of the first camera; $+X$ is camera right, $+Y$ horizontal forward, and $+Z$ gravity up. For entities visible in at least one input frame, it stores a Chinese category, 0.5 m-quantized center and extent, first/last evidence views, and a visible or seen_not_current status. Never-observed entities are excluded. This state is a separately requested auxiliary target, not an input to ordinary QA.

Typed operations define, but do not reveal, composition. Each question is normalized to a directed acyclic program over grounding G, frame alignment F, belief update B, metric estimation M, relation R, perspective P, and verification V. Node inputs and outputs are typed; the IR contains no answer value and remains hidden. The split withholds the full signatures counterfactual_cross_view.v1 and target_view_prediction.v1, while all seven atoms occur in training inside other programs. Thus the held-out target is a new composition of seen operations on unseen scenes, not an unseen operation token.

4 The EpiSpace Data Engine

4.1 Acquisition contract and raw-channel isolation

EpiSpace consumes existing OmniGibson acquisition bundles without mutating them. Each bundle contains a trajectory plan, ordered observations, camera transforms, an entity inventory, RGB/depth/instance sensor arrays, and quality reports. The sweep plan is the

Table 1: Channel isolation. Only the first row is used by primary SFT and benchmark inference.

Channel	Contents	Function
Model-visible	ordered raw RGB, camera height/FOV, natural language	ordinary multimodal reasoning; no pose, depth, IDs, program, or target render
Supervision-only	observable belief, typed answer-free program	optional state target, legality audit, signature label
Oracle-only	pose, metric depth, instance masks, world geometry, certificate	deterministic construction, margins, replay, held-out verification

Table 2: Pilot trajectory assets after strict typed quality gating. T10 RGB is held out and used only as a perspective verifier.

Class	Planned	Strict	Write bundles	Role
T1	46	28	28	write source
T3	92	83	83	write source
T4	46	10	10	write source
T7	46	43	43	write source
T8	46	8	8	write source
T10	46	21	0	verifier only

authoritative job ledger. The frozen policy accepts only jobs marked passed; the adapter additionally requires all core files, contiguous view indices, non-empty entities, integrity pass, and a compatible trajectory class. Review and failed jobs are preserved in a reason-coded ledger.

Human previews concatenate RGB, depth, and instance-ID panels and are therefore forbidden as model input. The loader extracts the uint8 RGB array from each sensor archive into a release-owned PNG, records its dimensions and raw pixel SHA-256, and stores instance pixel counts in a sidecar used only during compilation. A purity gate decodes every referenced image, verifies 1024×1024 RGB content against the sidecar, and rejects paths under a preview directory. Target-view renders are checked against all SFT inputs by both path and decoded RGB hash.

Across seven sweeps, 322 planned jobs produce 193 strict bundles. Of these, 172 are write sources with 1063 raw RGB views; 21 T10 bundles are verifier sources only. A fixed hash ranking assigns all descendants of each scene to 32/7/7 train/validation/test scenes. No trajectory, sibling family, or target render crosses that boundary.

4.2 Trajectory semantics and question legality

Camera motion determines which supervision is identifiable.

- T1 walkthroughs support overlap, revisit, separated-view registration, memory, metric, relation, and perspective reads.
- T3 rotation stations change yaw at a fixed position. We supervise two-frame change detection only when exactly one eligible object enters view; metric tasks are rejected because rotation adds no translational baseline.
- T4 adaptive arcs keep a focus instance visible across a collision-free partial orbit. The safe target is observed-instance identity, not canonical front or an unseen backside; these uniformly positive items are training-only.
- T7 elevation ladders change height and pitch at a station. Two sufficiently visible, unique objects are compared in an explicitly declared anchor-camera frame.

- T8 reveal paths provide a decisive category-presence view and matched evidence-deletion controls when such a family is valid.
- T10 target views never enter a write prefix. Their RGB is an oracle-only observation used to grade visibility predicted from a same-scene write trajectory.

T2, T5, T6, and T9 remain expansion slots and are not counted as pilot assets.

Source questions must have an accepted/unknown status, a terminal source certificate, and evidence views inside the actual model sequence. Instance questions are retained only when category-plus-view resolves one entity and at least one declared anchor contributes 500 instance pixels. T10 origin, facing, and candidate entities require their own unique observed anchors. Direction surfaces explicitly name the frame. All 241 observed ontology categories are deterministically localized into Chinese; an audit searches model-visible text for residual raw labels.

4.3 Geometry first, language second

The compiler fixes an entity binding and structured target before realizing a question. Relations are computed in the declared camera frame; distances are rounded only after metric evaluation; last-seen labels are recomputed over the actual presented view list; and unknown means that the required category or referent is absent from those inputs, not absent from simulator world truth. Natural-language templates then verbalize the fixed target. This ordering prevents a language model from inventing the geometry. Future paraphrasers must consume only question-side facts and remain answer-blind.

Each accepted item receives a globally unique fact ID, exact model/evidence views, entity IDs in hidden metadata, a typed program, a Chinese surface answer, and a replay certificate. The independent replayer does not trust a stored passed bit: it loads the source bundle and recomputes the target for all 17 task/program types. The frozen release replays 1093/1093 targets successfully.

4.4 Counterfactual family compiler

The generic family compiler expands validated atomic facts rather than relying only on rare hand-authored trajectories.

- Presence evidence (85 triples): one background RGB is shared by all siblings; the second RGB is either a category-free distractor, a $\geq$ 500-pixel decisive view, or a different distractor.
- Cross-view evidence (5 triples): two RGBs, including the canonical anchor and one resolvable referent, are shared; only the third image reveals or withholds the second never-co-visible referent.
- Trajectory reveal (1 triple): a T8 decisive view is exchanged against a matched category-free view.
- Frame equivariance (5 pairs): the same two images and objects are retained, but forward is anchored to image one versus image two; only pairs whose oracle relation changes are kept.
- Claim verification (37 pairs): each certified cross-view relation yields balanced true and false claims with a structured corrected relation.

This produces 133 complete families. Missing siblings, unequal image counts, inconsistent family IDs, and cross-split membership are release-blocking errors.

4.5 Exports and fail-closed validation

One intermediate episode IR is rendered into observe-first SFT, paired isolated SFT, optional observable-state SFT, RLVR-ready prompts, and benchmark rows. The full diagnostic benchmark contains 370 records. The recommended core is the union of all scene-disjoint family rows and all semantic-signature holdouts: 115 unique rows, with separate family and composition views for disaggregated reporting.

Eighteen release gates cover program typing; atom-seen/signature-unseen composition; scene and family locks; complete sibling shapes; shared family IDs; identical episode/isolated fact multisets; globally unique fact and record IDs; target-view path and hash isolation; raw-RGB purity; functional input–target dependence; terminal certificates; passing certificate checks; and full independent replay. All pass. The manifest also retains 677 exclusions, dominated by ambiguous references, failed or review acquisition jobs, trajectory-illegal tasks, small referents, and verifier-only bundles. Exclusions are part of the reported yield rather than silently discarded data.

5 Learning and Evaluation Interfaces

The student is never trained to print the hidden operation graph. All primary records use a Qwen-compatible multimodal chat schema: raw RGB and Chinese text are user content, and loss is applied only to assistant tokens. Hard spatial reads receive a short evidence-grounded explanation ending in the surface answer; simple presence and metric reads use the answer directly.

B: observe-first episode SFT. For each selected view set, the user first supplies all ordered images and then a numbered batch of at most six questions. The assistant returns aligned numbered answers. The frozen train arm contains 340 records covering 651 facts. With assistant mask m_t and per-record weight w_i, ordinary conditional SFT is

$$\mathcal{L}_B(\theta) = -\frac{\sum_i w_i \sum_t m_{it} \log p_\theta(x_{it} \mid x_{i,<t}, O_i)}{\sum_i w_i \sum_t m_{it}}. \tag{3}$$

System, user, image-placeholder, program, and certificate tokens have zero mask.

C: paired isolated-QA control. Each selected fact is separately exported over the same view list, yielding 651 records. A comparison ID links each isolated row to its episode record; explicit fact IDs prove equality of the two fact multisets. This gives two valid but different experiments.

In the fact-matched regime, each of the 651 facts has effective weight one. Episode presentation uses 340 draws and 1,168 image occurrences, whereas isolated QA uses 651 draws and 3,227 occurrences: episode structure saves 63.81% of image presentations by amortizing observations. This regime tests sample and inference efficiency, not equal visual compute.

In the stricter image-occurrence-matched regime, a deterministic sampler repeats an episode containing k selected facts exactly k times and assigns each repeat weight $1/k$. Both arms then contain 651 draws, 3,227 image occurrences, and 3,383,754,752 input pixels. Episode repeats contain 2,083 raw fact occurrences, but inverse-repeat weighting restores effective weight one for each of the same 651 facts. The schedule manifest stores source and output SHA-256 hashes, counts, pixels, a transparent image-token proxy, and every required invariant. Rebuilding in a clean directory is byte-identical.

D: observable-state auxiliary. The compiler exports 116 separate prompts that explicitly ask for the anchor-camera observable belief. Only observed entities and evidence indices may appear. D is an optional intervention, never a hint inside B or C. The causal study must first compare B and C alone, then add D at a reported ratio and test whether state-only answers remain correct after the RGB is removed.

RLVR-ready, not RL-completed. There are 651 prompts carrying hidden reward specifications for boolean, numeric-tolerance, categorical, structured-claim, or abstention targets. Family rows share a consistency identifier, enabling a future group reward after atomic answers are parsed. The artifact is verifier-ready; no claim is made that GRPO, policy optimization, or family reward training has been run.

Core benchmark protocol. The 115-row core is the union of 112 family rows and 27 signature-holdout rows, de-duplicated by record ID. A prediction file contains record_id and structured answer_value. The evaluator reports atomic accuracy by task/program/split, family exact match, evidence triple exact match, Eq. 2, balanced claim-pair exact match, frame-pair equivariance, and missing/extra predictions. Oracle smoke predictions recover every record and every family; this validates the evaluator wiring, not model capability.

Falsifiable model study. The primary future experiment trains the same 8B MLLM initialization under B and C using the image-occurrence-matched schedule and identical optimizer, general-data replay, and number of updates. The pre-specified primary outcome is scene-disjoint accuracy on the two held-out program signatures; co-primary diagnostics are family exact match and conditioned evidence sensitivity. Confidence intervals and hypothesis tests cluster by scene/family. State auxiliary and RLVR are subsequent ablations. A result in which B does not improve held-out composition under equal compute falsifies the episode-structure hypothesis even if training accuracy rises.

6 Data-Asset Validation

This section reports measurements of the frozen corpus, compiler, schedules, and evaluator. It contains no trained-model result. In particular, an oracle prediction file used below tests serialization and scoring code; it is not a spatial-reasoning baseline.

6.1 Frozen pilot yield

The acquisition ledger contains 322 terminated jobs over seven sweeps. The fail-closed loader retains 193 strict bundles: 172 write trajectories from T1, T3, T4, T7, and T8, plus 21 T10 bundles used only as target-view verifiers. The write trajectories span 46 scenes and 1,063 unique RGB views. Compilation produces 1,093 questions over 17 task/program types. The largest rejection source is ambiguous entity reference (422 cases); failed jobs (69), review jobs (60), trajectory–task incompatibility (90), sub-threshold referents (15), and verifier-only records (21) remain explicitly logged.

The training export contains 340 observe-first records, 651 paired isolated records, 116 optional observable-state records, and 651 verifier-ready prompts. The phrase verifier-ready describes a file format; no policy optimization has been run. All 133 registered groups have exactly their required sibling shape: 91 evidence triples (273 members), 37 claim pairs (74), and five frame pairs (10). Every group remains inside one scene split. All evidence families pass both set-matched single replacement and ordered-slot stability, so only the decisive slot changes. All frame pairs use identical RGB sequences across siblings and require different oracle relation labels.

6.2 Benchmark partitions and shortcut visibility

The full validation/test diagnostic export has 370 rows. The family view has 112 rows from 42 scene-disjoint sibling groups; the signature-composition view has 27 rows from the two program IDs withheld from training. Their de-duplicated union is the recommended 115-row core. The same-partition, task-conditioned majority oracle is 61.89%/62.50%/29.63%/62.61% on the full/family/composition/core views. It selects each task's most frequent evaluation target and is therefore a deliberately optimistic corpus diagnostic, not a learned no-image baseline. Its high family/core value is caused by explicit unknown siblings; family exact match and conditioned evidence sensitivity, rather than aggregate accuracy alone, are primary diagnostics.

The structured evaluator rejects missing, duplicate, extra, and malformed predictions as incorrect. Feeding it target-copied oracle predictions recovers 115/115 core rows, all 42 consistency groups, 28 test-time evidence triples, 12 claim pairs, and two frame pairs. This establishes that targets, grouping, numeric tolerances, and zero-denominator handling are wired consistently; it does not estimate VLM quality.

6.3 Independent integrity evidence

The builder and the serialized-corpus auditor exercise different trust paths. The builder independently re-executes every accepted certificate from the source geometry; the auditor reads only exported JSONL and images and does not import the compiler or verifier. Completed checks reproduce 1,093/1,093 targets; find all 133 families complete and all 91 evidence interventions ordered-slot stable; and find zero cross-split scene, trajectory, or family clusters. All 1,063 unique paths open as 1024×1024 RGB, and 20,503 model-visible text fields contain zero raw-ontology hits. Episode and isolated exports have identical 651-fact multisets. The two held-out program IDs are absent from training while every G, F, B, M, R, P, V atom is present; target-view path/hash isolation, identifier uniqueness, and normalized-input functionality also pass. The audit intentionally leaves model-dependent evidence sensitivity as N/A when no prediction file is supplied. Appendix B enumerates the eighteen release-blocking gates.

All image references are currently absolute paths to the release-owned RGB cache and therefore require path remapping when the corpus is moved. This is a portability limitation, not evidence leakage: the referenced files are raw RGB and no preview, depth, instance, pose, certificate, or oracle target render is model-visible.

6.4 Facts and visual compute are separate controls

The raw paired exports intentionally expose the information-structure difference. Under fact matching, both arms supervise the same 651 facts, but the episode arm presents only 1,168 images versus 3,227 for isolated QA, a 63.81% reduction through observation reuse. This is an efficiency comparison, not an equal-compute comparison.

For the causal model study, the deterministic image-occurrence-matched sampler produces 651 draws per arm and exactly 3,227 image occurrences (3,383,754,752 input pixels) per arm. Repeated episode records contain 2,083 raw fact occurrences; inverse-repeat weights restore effective weight one for each of the same 651 facts. Thus raw occurrence counts and effective supervision are reported separately.

Both schedule manifests pass their declared invariants and store source and output SHA-256 hashes. No claim about episode-learning gains follows from these data checks. The unexecuted, falsifiable comparison is the same 8B MLLM initialization trained under the two image-matched schedules, evaluated on the 115-row core with composition accuracy, family exact match, and Eq. 2 reported jointly.

7 Conclusion

EpiSpace turns spatial episode learning into a controlled empirical question. Its contribution is not another claim to simulator QA, executable programs, or cognitive maps in isolation. It is the conjunction of geometry-replayed counterfactual families, an atom-seen/signature-unseen composition split, and paired observe-first versus isolated training schedules with explicit fact and visual-compute accounting. The frozen pilot provides 1,093 certified questions, 133 complete sibling families, 651 paired training facts, and a 115-row core benchmark. Every target replays from source geometry, while model prompts contain only RGB and language. These results establish a validated data asset and evaluator, not that an MLLM has learned a better spatial representation. The decisive test remains the pre-specified matched training comparison.

Limitations. The pilot covers 46 BEHAVIOR-derived indoor scenes and is dominated by static observation; dynamic object state, interaction, and real-camera shift remain unmeasured. Eighty-five of 91 evidence triples concern category presence, so evidence intervention is broader than its present semantic diversity. The core has a high task-conditioned majority oracle because unknown siblings are explicit; joint family metrics are therefore mandatory. Media references are machine-local absolute paths and must be remapped after transfer. Deterministic Chinese realization favors geometric precision over linguistic breadth. Render media inherit the source academic-use terms. Finally, we have not yet run model training, RLVR, external transfer, or a human answerability study; none is implied by the reported corpus checks.

Reproducibility statement

The repository contains the trajectory taxonomy, compiler, split policy, training exporters, rejection ledger, corpus gates, and a manifest from which all pilot statistics in this draft are generated. Rendered media inherit the BEHAVIOR-1K academic-use boundary and are not claimed to be redistributable; code, schemas, and derived metadata can be released separately.

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A Serialized Data Contracts

All files are UTF-8 JSONL except manifests and reports, which are JSON. Table 3 lists the minimum public contract; additional hidden fields retain provenance and permit audits.

Table 3: Principal serialized artifacts.

Artifact	Required content
episodes.ir.jsonl	episode/scene/trajectory IDs, split lock, ordered observations, observable belief, and question list
train.episode_sft.jsonl	messages, assistant-only loss policy, and comparison contract for an observe-first question batch
train.isolated_sft.jsonl	one paired fact per dialogue with the same evidence views and comparison ID
train.state_aux_sft.jsonl	separately requested observable-state target; never inserted into primary QA
train.rlvr.jsonl	prompt and hidden deterministic reward specification; no claim that policy training was run
benchmark*.jsonl	record/family/scene IDs, split, model input, structured target, hidden program, certificate, and evidence provenance

An IR observation is {view_id, step, role, rgb}. A question stores fact_id, task, Chinese surface question/answer, structured answer value/status, program, model/evidence view IDs, family variant, and certificate. The answer-free program contains a semantic signature, typed atoms and nodes, external inputs, and one answer node. The certificate contains named checks, oracle-only replay inputs, terminal result, and verifier version. Programs, certificates, entity IDs, pose, depth, and masks are excluded from ordinary model messages.

Episode and isolated records share a comparison_id; their comparison contracts declare fact IDs, unique-image count, question count, and arm. The minimal prediction format is:

{"record_id":"benchmark-...","answer_value":<JSON value>}

The evaluator requires exactly one prediction per benchmark record. Strings, booleans, nulls, lists, and dictionaries are compared with exact types and keys; finite numeric fields use the tolerance stored in the certificate.

B Fail-Closed Gates

The release manifest records eighteen Boolean gates. A failed gate makes the build fail rather than merely attaching a warning:

1. typed_programs_valid, heldout_program_absent_from_train, and heldout_atoms_seen_in_train;
2. scene_split_lock, family_split_lock, family_members_complete, and family_intervention_contract;
3. family_id_shared_within_consistency_group;
4. paired_fact_multiset_equal, fact_ids_globally_unique, and record_ids_unique;
5. heldout_target_rgb_absent_from_sft_input, heldout_target_rgb_hash_absent_from_sft_input, rgb_channel_purity, and normalized_input_functional_dependency;
6. all_certificates_terminal, all_certificate_checks_pass, and certificate_full_reexecution.

The independent corpus audit is intentionally separate. It parses every serialized row, recomputes family shapes and cluster leakage, opens all image paths, scans model-visible text for raw ontology tokens, measures target priors, and compares episode/isolated facts and image exposure. It never calls the builder, compiler, exporter, or geometry verifier. Without a prediction file, model-dependent evidence sensitivity is reported as not evaluated.

C Metrics

Let c_i be structured exact correctness for record i. Atomic accuracy is the mean of c_i, disaggregated by task, program, and split. For a sibling family f, family exact match is $\prod_{i \in f} c_i$. Evidence-triple exact match and the conditioned score in Eq. 2 are both required: reveal accuracy is the denominator, so an always-unknown model cannot receive evidence-sensitivity credit. Claim-pair exact match requires both true and false siblings. Frame equivariance requires both same-RGB frame answers to match their respective targets; merely changing between two wrong labels earns no credit. Composition accuracy is atomic accuracy on the two held-out semantic signatures and is always reported separately from family metrics.

Missing, duplicate, extra, malformed, or non-finite predictions are incorrect. When model results are eventually produced, uncertainty is computed by resampling the scene/family unit rather than treating sibling rows as independent. The same-partition task-majority number is labeled an oracle diagnostic and is never presented as a trained no-image baseline.

D Matching Regimes

The fact-matched schedules require equality of the raw and effective fact multisets. They preserve the natural amortization difference: 340 episode draws/1,168 image occurrences versus 651 isolated draws/3,227 occurrences. The image-occurrence-matched schedules repeat an episode once per selected fact and assign each repeat inverse multiplicity. Required invariants are equality of the effective fact-weight multiset, exact image-occurrence multiset, and pixel exposure. The resulting two arms each contain 651 draws, 3,227 image occurrences, and 3,383,754,752 pixels. Text-token counts are a Unicode-lexeme proxy and are explicitly not claimed to equal a model tokenizer.

E Reproduction Commands

Commands are run from the repository root with the project interpreter.

```
PY=../habitat/.conda/core/bin/python

$PY -m episode3d.cli \
  --config configs/epispace_pilot_v1.json

$PY scripts/build_compute_matched_schedules.py \
  --episode data/epispace_pilot_v1/train.episode_sft.jsonl \
  --isolated data/epispace_pilot_v1/train.isolated_sft.jsonl \
  --output-dir data/epispace_pilot_v1/compute_matching \
  --seed 17

$PY -m episode3d.corpus_audit \
  data/epispace_pilot_v1

$PY -m episode3d.benchmark_evaluator oracle \
  --benchmark data/epispace_pilot_v1/benchmark.core.jsonl \
  --output /tmp/epispace-oracle.jsonl

$PY -m episode3d.benchmark_evaluator evaluate \
  --benchmark data/epispace_pilot_v1/benchmark.core.jsonl \
  --predictions /tmp/epispace-oracle.jsonl \
  --output-dir /tmp/epispace-evaluation

$PY -m pytest -q
$PY -m ruff check episode3d scripts tests
```

The oracle command copies structured targets solely for evaluator smoke testing. It must never be used as a model result. The authoritative frozen statistics are in release_manifest.json; serialized-only diagnostics are in corpus_audit.json; schedule invariants and hashes are in compute_matching/compute_matching_manifest.json.

F Asset and License Boundaries

Raw RGB PNGs are extracted into a release-owned cache, but current JSONL rows store machine-local absolute paths. Consumers must remap the cache prefix without changing content hashes. BEHAVIOR-1K-derived renders remain governed by their source academic-use terms. The present internal artifact does not grant redistribution permission. No real-world transfer, human calibration, or model-training result is part of this frozen release.